Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	LTVIP2026TMID00420
Project Name	EV Market Analysis Using Tableau
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python, SQL HTML, CSS, etc	Python is used for cleaning 17M rows Vahan data, handling nulls & filtering EV data only
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	Snowflake / MySQL / Excel / CSV	CSV/Excel is used to store cleaned Vahan Parivahan EV dataset for Tableau import
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Snowflake Stages / COPY INTO / Tableau Data Connections	To load EV data from Kaggle (Vahan dataset) directly into Tableau Desktop
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Eg. Tableau prep, Snowflake SQL, Tableau Desktop	To clean data, create KPIs like Total Sales, YoY Growth, Market Share, Top Makers
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Snowflake SQl Charts, Tableau , Python	To create interactive dashboards - Map, Bar, Line, Pie charts for EV analysis
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public, Flask,Streamlit	User can filter by State, Year, Category & view Story with 5 slides of insights